

Course- B.Tech.	Type- Core
Course Code- 2025CSET106	Course Name- Discrete Mathematical Structures
Year- I	Semester- Even (2 nd Semester)

Tutorial- 11

Tutorial No.	Name	CO1	CO 2	CO3
11	Pigeonhole Principle, Recurrence Relations and Generating Functions	✓	--	✓

Objective: The main objective of this tutorial is to strengthen the understanding of the pigeonhole principle, solving recurrence relations, and using generating functions through simple exercise-based questions.

Questions:

1. Show that among any n consecutive integers, there is exactly one integer divisible by n .
2. Show that if five integers are selected from the first eight positive integers, then some two of them have sum 9.
3. Find the generating function for the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$, for $n \geq 2$, with initial conditions $a_0 = 1$ and $a_1 = 0$.
4. Find the generating function for the recurrence relation $a_n = a_{n-2}$, for $n \geq 2$, with initial conditions $a_0 = 5$ and $a_1 = -1$.
5. Find the generating function for the finite sequence 1, 4, 16, 64, 256.
6. Find a closed form for the generating function of the sequence 2, -2, 2, -2, 2, -2,
7. How many solutions does the equation $a_1 + a_2 + a_3 = 17$ have if $2 \leq a_1 \leq 5$, $3 \leq a_2 \leq 6$ and $4 \leq a_3 \leq 7$?